Chapter 6
Schottky Diodes

Schottky diodes are formed by a metal-semiconductor junction, whose rectifying behavior was described early [Sch38, Sch39]. They are unipolar devices, which means that only one type of carriers is available for the current transport. If they are designed for large blocking voltages, the resistance of the base will increase strongly due to the lack of charge carrier modulation. Schottky power diodes have been used for a long time, but in the last years they have gained an increased importance in following areas:

- Si Schottky diodes in the voltage range up to approximately 100 V to be used as freewheeling diodes for MOSFETs. Their advantage is the low junction voltage and the absence of stored charge. At turn-off from conducting to blocking mode, only the capacitive recharge of the junction capacitance needs to be considered. This makes them very useful for very high switching frequencies.
- Schottky diodes from semiconductor materials with wide bandgap. With these materials, much higher blocking voltages are possible due to the higher critical fields. Because of the much smaller possible junction voltage compared with wide-gap pn-diodes, Schottky diodes have become an attractive alternative.

6.1 Energy Band Diagram of the Metal-Semiconductor Junction

Figure 6.1 shows the band structure of a contact between a metal and an n-type semiconductor. We focus to this case which because of the higher mobility of electrons than of holes is solely used for power Schottky diodes. An ideal contact with negligible influence of surface states is considered. The diagram is determined by the work functions Φ_M and Φ_S of the metal and the semiconductor and the electron affinity χ of the semiconductor. The work function of a body not in contact with others is defined as the difference between the vacuum level, i.e. the energy far

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J. Lutz et al., *Semiconductor Power Devices*,
https://doi.org/10.1007/978-3-319-70917-8_6

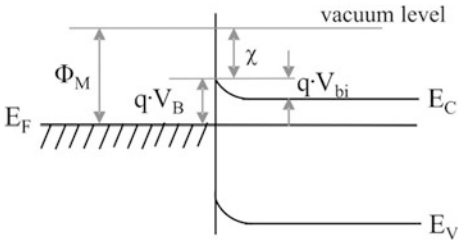


Fig. 6.1 Band diagram of a metal-semiconductor junction in thermal equilibrium. Surface charges are neglected. Figure according to [Spe65]

outside, and the Fermi level; the electron affinity χ of a semiconductor is the difference between the vacuum level and the conduction band edge. Since in a non-degenerate semiconductor the Fermi level is below the conduction band edge, χ is smaller than the work function Φ_S. If now a metal and an n-type semiconductor with $\Phi_S < \Phi_M$ are put into contact, the Fermi level in the semiconductor is initially higher than in the metal, since the vacuum level remains constant (the space outside is field-free in thermal equilibrium). Hence electrons flow from the semiconductor into the metal leaving a positive space charge region in the semiconductor formed by depleted donors. This is connected with a bend-up of the bands and continues until equilibrium is reached and the Fermi level is constant throughout the structure, see Fig. 6.1. At the interface then the following energy step occurs:

$$qV_B = \Phi_M - \chi \tag{6.1}$$

This barrier, which electrons moving from the metal to the semiconductor have to overcome, is called contact barrier or Schottky barrier. In the semiconductor, the bend-up of the bands means that a built-in voltage V_{bi} is formed over the space charge region:

$$qV_{bi} = \Phi_M - \Phi_S = \Phi_M - (\chi + (E_C - E_F)) \tag{6.2}$$

Compared with the isolated semiconductor the electron energy in the neutral part of the n region is reduced by qV_{bi}, the neutral n region is on a positive potential compared to the metal. Although electrons moving from the semiconductor to the metal have to surmount a smaller barrier (qV_{bi}) than electron moving from the metal to the semiconductor, the two currents compensate each other in thermal equilibrium, since the relevant carrier density in the conduction band due to its distance from the Fermi level is smaller than in the metal.

The Eqs. (6.1) and (6.2) hold also if $\Phi_M < \Phi_S$. Then no depletion layer is formed in the semiconductor, but instead an enhancement layer of electrons. According to (6.2) the built-in voltage is negative then. For $\Phi_M = \chi$ the Schottky barrier qV_B vanishes after (6.1). In this case, the contact behaves like an ohmic contact, since the contact resistance $R_c = (dj/dV)^{-1}$ is very small (see [Sze81], p. 304). Usual *ohmic contacts* differ from Schottky contacts (i) by a large number of interface states with levels in the bandgap resulting in a high surface recombination velocity, which in Schottky junctions is negligible; (ii) usual ohmic contacts are formed with

a highly doped semiconductor with a doping $> 5 \times 10^{18}$ cm^{-3}. In this range the barrier is so thin that charge carriers can easily tunnel through.

6.2 Current-Voltage-Characteristics of the Schottky Junction

If a voltage V is applied to the metal relative to the semiconductor, the potential difference V_{bi} between the semiconductor and the metal is reduced for $V > 0$ and enhanced for $V < 0$. This is connected with a narrowing or widening of the space charge region in the semiconductor, respectively. The barrier for the electrons flowing from the semiconductor to the metal is changed to $q(V_{bi} - V)$, whereas the contact barrier qV_B to be surmounted by electrons flowing from the metal to the semiconductor remains constant. Hence for $V > 0$ the electron current flowing from the semiconductor to the metal soon predominates strongly, the junction is conducting. For $V < 0$, the voltage-independent electron current flowing from the metal to the semiconductor predominates, and since this current is small due to the higher barrier qV_B, the junction is blocking in this direction. Assuming that the electrons obey the Boltzmann distribution and both the currents, from the semiconductor to the metal and from the metal to the semiconductor, are proportional to the electron concentration at the interface, the I–V characteristic is obtained as [Sze81]:

$$j = j_s \cdot \left(e^{\frac{q \cdot V}{k \cdot T}} - 1 \right) \tag{6.3}$$

In this general form the equation is identic with Eq. (3.50) of the ideal pn-junction, see also Fig. 3.10. However the saturation current density j_s is quite different, namely given by

$$j_s = A^* \cdot T^2 \cdot e^{-\frac{q \cdot V_B}{kT}} \tag{6.4}$$

where qV_B is the contact barrier defined in (6.1) and A^* a semiconductor specific constant, the so called effective Richardson constant. The characteristic is independent of the bandgap and doping density of the semiconductor. For a metal on n-semiconductor contact, V is the voltage applied to the metal relative to the semiconductor, for a metal contact on a p-semiconductor the polarity is inverse.

The effective Richardson constant for some semiconductors is given in Table 6.1. In the inset table of Fig. 6.2, the contact barriers of important contact metals for Si Schottky diodes are compiled. Mainly silicides are used for the contact, since these facilitate tuning and optimization of the contact properties. The saturation current densities calculated by Eq. (6.4) are given also, to show the very strong increase with decreasing barrier height. j_s is in all cases several orders of magnitude higher than for pn-junctions in Si. The shown I–V characteristics obtained with these data illustrate on the other hand that the threshold voltage

decreases with decreasing V_B. That the threshold voltage for listed contact metals is significantly smaller than for pn-junctions, is the reason why Schottky junctions are applied at all.

Similar to pn-junctions the saturation current increases strongly with temperature. For Cr_2Si for instance the threshold voltage (if defined as the voltage at 5 A/cm^2) is only 0.2 V, but the leakage current is about 2 mA/cm^2 at 300 K and about 1 A/cm^2 at 400 K. The Cr_2Si Schottky contact is therefore only suited for very low blocking voltages, e.g. for diodes in switched mode power supplies for low voltages. For 100 V Schottky diodes, typically PtSi is used as contact material.

In blocking direction the ideal I–V characteristics according to (6.3), (6.4) is observed only for relatively small voltages. For larger reverse biasing the blocking current is enhanced by the image force [Sze81]. This leads to a lowering of the contact potential by

$$\Delta\Phi = \sqrt{\frac{q \cdot \mathsf{E}}{4 \cdot \pi \cdot \varepsilon}} \tag{6.5}$$

where E is the field strength at the contact. Hence instead of (6.4) the saturation current density at higher reverse voltage is given by:

$$j_s = A^* \cdot T^2 \cdot e^{-\frac{q \cdot (V_B - \Delta\Phi)}{kT}} \tag{6.6}$$

Table 6.1 Effective Richardson constant A^* for different semiconductors of n-type

Si	110 A/(cm^{2}K^2)	[Sze81]
GaAs	8 A/(cm^{2}K^2)	[Sze81]
SiC	400 A/(cm^{2}K^2)	[Tre01]

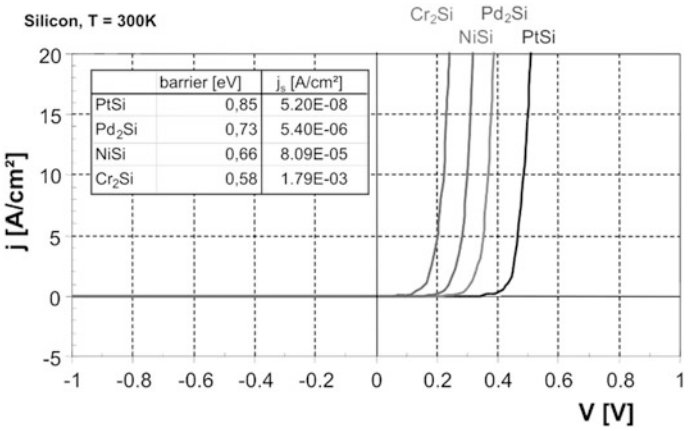


Fig. 6.2 Calculated I-V characteristics for Si Schottky diodes for different contact materials according to data from berndes, IXYS semiconductor GmbH [Ber97]. The characteristics and saturation current densities in the inset refer to 300 K

The increase of $\Delta\Phi$ with the applied reverse voltage is the main reason for the "soft" reverse characteristics of Schottky diodes. The effect becomes especially noticeable for Schottky diodes with low barrier heights or Schottky contacts on wide-gap semiconductors. At $\mathsf{E} = 2 \times 10^5$ V/cm the barrier lowering amounts to 50 meV.

In forward direction, Eq. (6.3) is valid at higher current densities only, if the voltage V is defined as the voltage across the junction, excluding the resistive voltage drop over the neutral semiconductor. The whole forward voltage as function of j is given at higher current densities as the sum of the junction voltage obtained from (6.3) and the ohmic voltage drop:

$$V_F = \frac{kT}{q} \ln\left(\frac{j}{j_s} + 1\right) + r_\Omega j \tag{6.7}$$

where $r_\Omega = R_\Omega \cdot A$ is the resistance times active area A. Since the resistance of unipolar devices is constant, the I–V characteristic is completely given by (6.7). Because conductivity modulation does not take place, the resistance is much larger than for a pin diode with equal breakdown voltage. R_Ω is called also on-resistance, since relevant only for the forward voltage or the on-state. In Sect. 6.4 R_Ω is discussed in detail.

Also an ideality factor n is sometimes introduced in the exponent of Eq. (6.3) to obtain a better agreement with experimentally measured values:

$$j = j_s \cdot \left(e^{\frac{q \cdot V}{n \cdot k \cdot T}} - 1\right) \tag{6.6b}$$

The value for n is between 1 and 2, but for good Schottky diodes a value of 1.02–1.06 can be obtained. The main reasons for a non-ideal behavior, i.e. $n \neq 1$, are states at the interface.

6.3 Structure of Schottky Diodes

Figure 6.3 shows the basic structure of a Schottky diode. The low doped n^- layer, which must sustain the reverse voltage, is grown on the n^+ substrate by epitaxy. Real Schottky diodes need additional junction termination structures, not shown in the figure. Commonly used terminations are field plates, the diffusion of potential rings, or a JTE-structure. Sometimes also combinations of field plate and potential ring or JTE-structures are implemented. For details of these junction terminations, see Chap. 4.

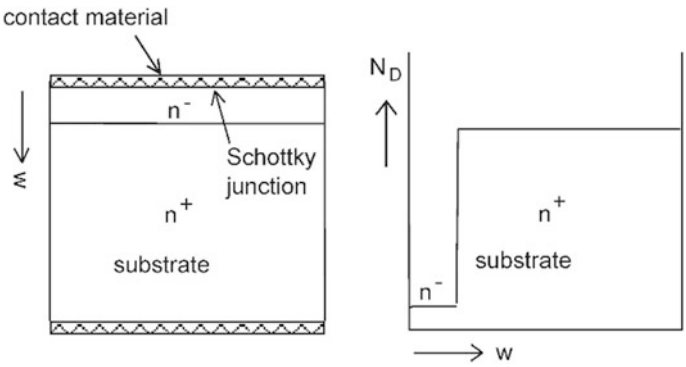


Fig. 6.3 Basic structure of a Schottky diode

6.4 Ohmic Voltage Drop of a Unipolar Device

The resistance of the low-doped middle layer in a unipolar device is given by the following expression:

$$R_\Omega = \frac{w_B}{q \cdot \mu_n \cdot N_D \cdot A} \tag{6.8}$$

where as previously w_B denotes the thickness of the layer, N_D the doping concentration (carrier density), μ_n the electron mobility and A the active area of the device. Because with increasing blocking voltage the required base width increases and the allowed doping concentration decreases, the resistance increases strongly with breakdown voltage. For a triangular field distribution, the base width for a breakdown voltage V_{BD} is $w_B = 2V_{BD}/\mathsf{E}_c$, the maximal doping concentration according to (3.74) is $N_D = \varepsilon \mathsf{E}_c^2/(2qV_{BD})$, where E_c is the critical field of the semiconductor. Substituting this into (6.8) one obtains for the resistance times area.

$$r_\Omega = R_\Omega A = \frac{4V_{BD}^2}{\mu_n \mathsf{E}_c^3} \tag{6.9}$$

If the dependence of μ_n on N_D and hence on V_{BD} is left out of account, the resistance is proportional to the square of the breakdown voltage, the material parameter E_c enters with the third power. The latter is the cause why the on-resistances of unipolar devices of different semiconductors such as Si and SiC is extremely different. The critical field decreases slightly with breakdown voltage as given by Eq. (3.88). Inserting (3.88) one obtains:

$$r_\Omega = \frac{1}{\varepsilon \mu_n} \left(\frac{B}{n+1} \right)^{\frac{3}{n-1}} (2V_{BD})^{\frac{2n+1}{n-1}} \tag{6.10}$$

Due to the decrease of E_c the resistance increases yet stronger than with the square of V_{BD}. This holds even taking the increase of μ_n with V_{BD} into account. The mobility is obtained inserting the doping concentration given by (3.87) as function of the voltage into the formula (2.35) using the parameters of Appendix A. If the real base width is equal to the unhindered extent of the space charge region (NPT-case), the resistance is necessarily given by (6.8) and, as far as (3.88) is applicable, by Eq. (6.10).

Passing to PT-design with trapezoidal field shape, the resistance for a given V_{BD} changes, since w_B and N_D will decrease differently from their NPT-values. At moderate PT the resistance has a minimum, as can be calculated expressing N_D from (5.7) by w_B assuming E_c as independent of N_D and inserting it into (6.8). With μ_n assumed constant also, differentiation for constant V_{BD} yields that R_Ω is minimal at [Hu79]

$$w_B = \frac{3}{4} \left\{ \frac{2V_{BD}}{\mathsf{E}_c} \right\}, \qquad N_D = \frac{8}{9} \left\{ \frac{\varepsilon \mathsf{E}_c^2}{2\mathrm{qV_{BD}}} \right\} \tag{6.11}$$

This means that the field E_w at the nn^+-junction is one third of the critical field: $\mathsf{E}_w = \mathsf{E}_c/3$. With (6.11) in (6.8) the resistance is reduced to 27/32 = 0.844 of the value given by (6.9) for NPT-design.

The *dependence* of the resistance on N_D or w_B for a given breakdown voltage [Dah01] can be calculated solving Eq. (5.7) first for w_B which yields:

$$w_B = \frac{2V_{BD}}{\mathsf{E}_c + \sqrt{\mathsf{E}_c^2 - 2\frac{q}{\varepsilon} V_{BD} N_D}} \tag{6.12}$$

The equation holds up to the N_D-value for NPT, where the square root term vanishes. E_c depends on N_D according to Eq. (3.78)

$$\mathsf{E}_c = \left(\frac{q(n+1)N_D}{\varepsilon B} \right)^{\frac{1}{n+1}}$$

which holds also for moderate PT with $\mathsf{E}_w \lesssim \mathsf{E}_c/2$, see Sect. 5.3. Substituting this for E_c in (6.12) and inserting the equation into (6.8) one obtains the resistance as function of N_D for a given V_{BD}. The result for a Si Schottky diode with 240 V blocking voltage is shown in Fig. 6.4. Besides the resistance r_Ω also the base width is plotted versus N_D. For a good approximation of α_{eff} for this voltage the exponent n = 6 was used, which according to (3.83) leads to $B = 5.45 \times 10^{-30}$ $\mathrm{cm^{n-1}/V^n}$.

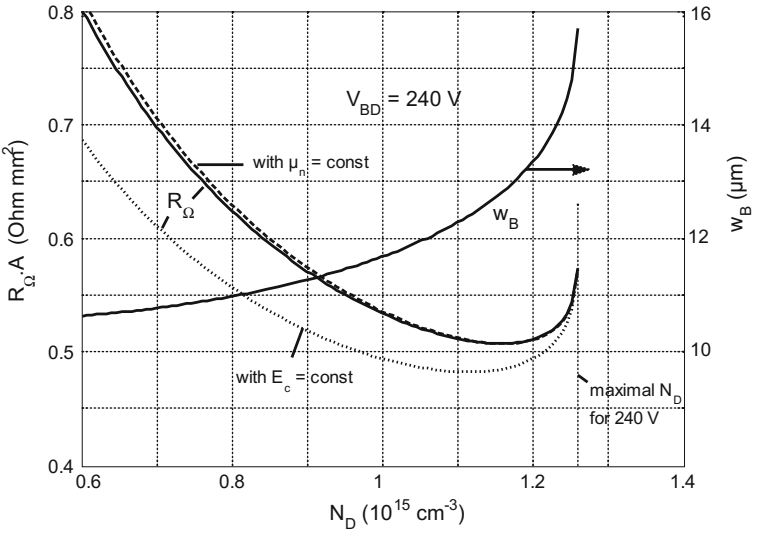


Fig. 6.4 Calculated resistance R_Ω and base-width w_B for a 240-V silicon Schottky diode in dependency of the base doping

In the first range below the maximal doping (NPT-doping), w_B decreases very strongly, thus overcompensating the decrease of N_D. For r_Ω three lines are shown: The solid line shows the exact result allowing not only for the variation of E_c but also the dependence of μ_n on N_D. The dashed curve is calculated with variable E_c but constant mobility μ_n using the value 1360 cm^2/(Vs) for the doping concentration at the minimum of R_Ω. As is seen, the variation of μ_n has not a significant effect in the considered range. For the dotted curve the critical field was assumed constant as for the Eq. (6.11) using the NPT-value given by (3.88) (μ_n being variable). As the Figure shows, the resistance obtained with this approach is clearly too small. According to the solid curve the lowest possible resistance of a 240 V-device is 0.508 Ω mm^2 = 0.89 times the resistance in NPT-layout. Furthermore, the N_D-value of minimal R_Ω is located closer to the N_D-value for NPT than according to (6.11).

This approach yields a considerably smaller resistance than more correctly calculated with variable E_c (solid curve). The calculation for other blocking voltages leads to similar results. This is supported by analytical expressions for the minimum resistance and the N_D-value of the minimum, taking the variation of E_c into account (in contrast to (6.11)). From Eqs. (5.7) and (3.78) one determines first the derivative dw_B/dN_D for constant V_{BD} and uses this for the differentiation of (6.8). The result of this somewhat lengthy calculation is that for minimal R_Ω the field strength E_w at the nn$^+$ junction in relation to the critical field E_c (at the metal semiconductor junction) must obey the condition:

$$\frac{\mathsf{E}_w}{\mathsf{E}_c} = \frac{n-1}{n+1}\,\frac{1}{3} \tag{6.13}$$

The field ratio varies from 1/3 for large n (constant E_c) over 1/4 for n = 7 to 1/5 for n = 4, an exponent value appropriate for 50 V-devices. The decrease of E_c with decreasing N_D moves the minimum nearer to NPT. From the known ratio $\mathsf{E}_w/\mathsf{E}_c$ one can determine the doping concentration and base width at the minimum of R_Ω and hence the minimal resistance itself. The results are in close agreement with Fig. 6.4. The factor between the minimum resistance and the NPT-value is nearly independent of the blocking voltage. For Si-devices the minimum resistance is (88 ± 2) % of the NPT-value. Hence using Eq. (6.10) with n = 7 one obtains for the minimal resistance of unipolar Si devices:

$$R_{\Omega,\min} = 0.88 \cdot \frac{2 \cdot B^{\frac{1}{2}} \cdot V_{BD}^{\frac{5}{2}}}{\mu_n \cdot \varepsilon \cdot A} \tag{6.14}$$

Figure 6.5 shows the minimal resistance and the appropriate width of the middle layer of a unipolar Si-device as a function of the breakdown voltage V_{BD}. The NPT-resistance was calculated from (6.10) with n = 6 for this voltage range and is multiplied then with the factor 0.88 as described above. The base width was calculated multiplying the NPT-value by 0.817, a factor derived from Eq. (6.13) together with (3.78) (see also Fig. 6.4). The resistance is growing in this range nearly with the power of 2.5 of the breakdown voltage. As a result, unipolar devices for high blocking voltage will have a very high on-state resistance.

For silicon Schottky diodes, often a deviation from the unipolar resistance behavior is observed. To achieve sufficient blocking capability, a p-doped potential ring is usually implemented at the edge of the active area, similar to Fig. 4.24 in Chap. 4. In this case a pn-junction is connected in parallel to the Schottky junction, and at a forward voltage in the range of the diffusion voltage of the pn-junction this junction starts to inject charge carriers which reduce the resistance. Because the

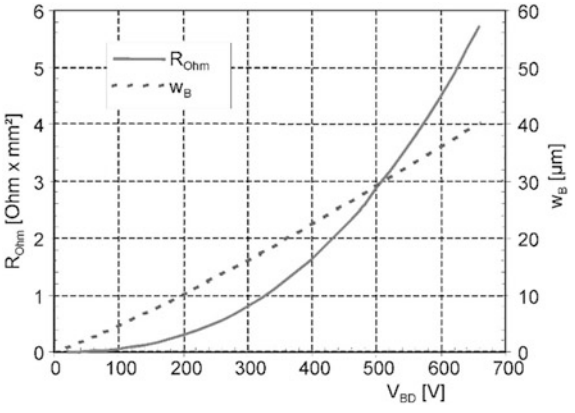


Fig. 6.5 Minimum resistance and appropriate width of the middle layer of a unipolar Si-device in dependence of breakdown voltage

injection increases with temperature owing to the decrease of the threshold voltage, a decrease of the resistance with temperature is found in such cases, whereas an increase is expected from the decrease of the mobility. For Schottky diodes based on SiC, the pn-junction of which begins later to inject, the resistance shows the temperature dependence as determined by the mobility μ_n. Details on this issue will be discussed below, see Fig. 6.9.

6.4.1 Comparison of Silicon Schottky Diodes and pin Diodes for Rated Voltages of 200 and 100 V

Because some spread of data is to be expected from tolerances in used materials and fabrication processing steps (roughly 10% as a role of thumb) and to have some safety for tolerances in the measurement technique (up to 10%), the calculation for a 200 V rated diode is done for a breakdown voltage of 240 V. From Eq. (6.15) a value for $R_{\Omega} \cdot A = 0.51\ \Omega.\text{mm}^2$ is obtained. At a current density of 1.5 A/mm^2 – a typical current density at rated current – a voltage drop across the epi region of 0.77 V results. As Schottky contact material PtSi can be used. This material leads to a threshold voltage of 0.5 V, and therefore a forward voltage $V_F = V_S + R_{\Omega} \cdot A \cdot j = 1.27$ V is to be expected. In practice, however, less than 0.9 V is measured. This discrepancy can be explained by the bipolar effect, as discussed above.

In comparison, a fast epitaxial pin diode designed for a rated voltage of 200 V can be fabricated so that a forward voltage smaller than 1 V is reached at the assumed current density, although the junction voltage is in the range 0.7 – 0.8 V. The stored charge to be extracted during commutation is comparable to the capacitive charge of the Schottky diode.

For a rated voltage of 100 V, however, the Schottky diode designed with similar tolerances shows a resistance $R_{\Omega} \cdot A = 0.082\ \Omega\ \text{mm}^2$. With the PtSi barrier again the resulting on-state voltage is $V_F = V_S + R_{\Omega} \cdot A \cdot j = 0.62$ V. Such a low value cannot be reached with a pin-diode. For still lower voltages, the advantage of the Schottky diode compared to a pin diode will be even higher.

6.5 Schottky Diodes Based on SiC

6.5.1 SiC Unipolar Diode Characteristics

The first SiC Schottky diodes were developed by the German company SiCED and introduced on the market by Infineon for rated voltages of 600 and 1200 V. SiC Schottky diodes are superior to solutions using bipolar Si-diodes in all applications that require a high switching frequency (switched mode power supplies, power factor correction, etc.) [Zve01].

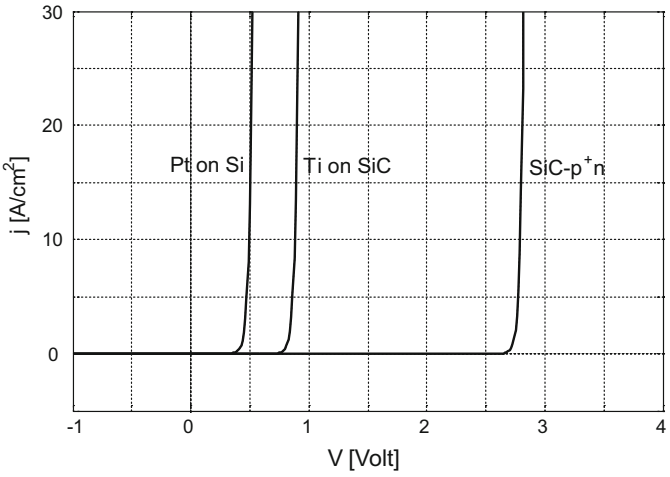


Fig. 6.6 Ideal low-current characteristics of different junctions based on SiC and Si

The used SiC polytype is 4H which has a bandgap of 3.26 eV. For a SiC pn-junction the high bandgap leads to a threshold voltage of about 2.7 V, as can be seen from Fig. 3.11. The high forward voltage drop is a clear disadvantage of SiC pn- and pin-diodes, which is why diodes in SiC are usually Schottky-diodes or Merged pin Schottky diodes. In Fig. 6.6 the forward characteristic of a SiC-Schottky diode with a Ti-contact, which has a contact barrier qV_B = 1.27 eV, is compared in a range of lower current densities with characteristics of a SiC pn-junction (parameters as in Fig. 3.11) and a Si-Schottky junction with PtSi-contact (qV_B = 0.85). The ideal characteristics are assumed with effective Richardson constants taken from Table 6.1. The advantage of the SiC-Schottky-junction against the pn-junction is obvious.

For Schottky-junctions on SiC typically metals with a higher contact barrier are used than for Si, because the higher field at reverse bias results in a strong lowering of the barrier by the image force. The Ti-contact is often used for high-voltage Schottky diodes.

One of the main advantages 4H-SiC compared to silicon is a well ten times higher critical field strength for avalanche breakdown. As has been expressed in Eq. (2.101), the effective ionization rate is found to obey the power dependency $\alpha_{eff} = B|\mathsf{E}|^n$ with n = 8.03 and $B = 2.185 \times 10^{-48}$ cm^{n-1}/V^n [Bar09]. On the base of these parameters, specified Eqs. (3.91) and (3.92) have been given for the critical field and breakdown voltage as functions of N_D. The latter relationship has been compared in Fig. 3.19 with that of silicon. Equations containing n and B explicitly such as (3.86–3.88) are of course also immediately applicable to SiC. At V_B = 1500 V Eq. (3.88) yields for the critical field of 4H-SiC $\mathsf{E}_c = 2.64\,MV/cm$, which is a factor 11.8 higher than obtained for Si with n = 7, $B = 2.107 \times 10^{-35}$ cm^{n-1}/V^n. This passes in a reciprocal manner to the base width $w_B = 2V_{BD}/\mathsf{E}_c$ which is by the same factor *smaller* than for a silicon device. A plot

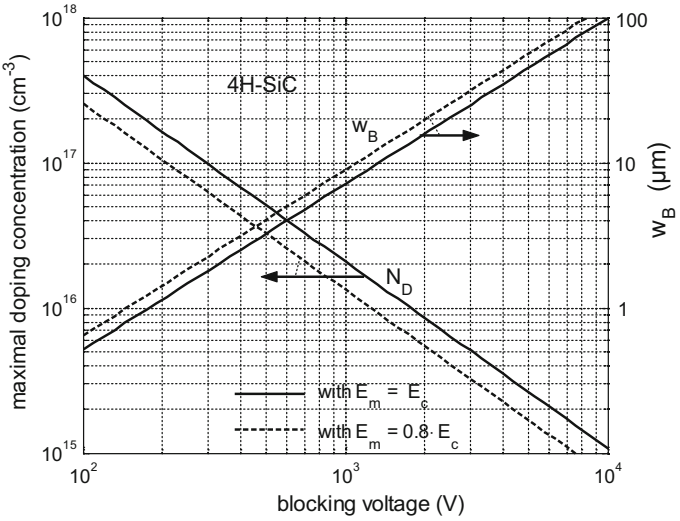


Fig. 6.7 Doping concentration and base width versus usable blocking voltage for a SiC-diode with triangular field distribution. For the solid lines the allowed maximum field E_m is put equal to the critical field E_c, for the dashed curves E_m is put equal to 0.8 E_c

of the doping concentration and base width versus the "usable" blocking voltage is shown in Fig. 6.7. The solid curves are obtained if the usable blocking voltage is equated with the breakdown voltage V_{BD} ($\equiv V_B$). For the dashed curves the usable blocking voltage is defined as the voltage at which the field reaches 80% of the critical field. This corresponds to practical guidelines for the design, which have to take into account that SiC-devices exhibit soft reverse characteristics with substantially increasing current appreciably below the breakdown voltage, as shown further below in Fig. 6.9. Therefore the field strength is limited to a value considerably below E_c. The respective curves in Fig. 6.7 were calculated replacing the critical field in $w_B = 2V_{BD}/E_c$ and $N_D = \varepsilon E_c^2/(2qV_{BD})$ by $0.8 \cdot E_c$ and substituting then E_c by (3.88). Even with such cutback, the necessary base width of SiC devices is nearly an order of magnitude smaller and the allowed doping density N_D two orders higher than for silicon (see Fig. 3.17).

Both inserted into (6.7) results in a unipolar on-resistance which is nearly three orders of magnitude smaller than for Si-devices. This follows already from Eq. (6.8) because of the one order higher critical field. For a definite calculation of R_Ω one needs the electron mobility. According to [Scr94] (see Appendix A2) the electron mobility in 4H-SiC depends at 300 K in the following manner on the doping density:

$$\mu_n = \frac{947}{1 + \left(\frac{N_D}{1.94 \cdot 10^{17} cm^{-3}}\right)^{0.61}} \frac{cm^2}{Vs} \tag{6.15}$$

μ_n is somewhat smaller than in Si, and since the doping of SiC-devices is higher, its decrease with increasing doping (decreasing V_{BD}) comes stronger into play. Substituting N_D in (6.15) by (3.87) and inserting the obtained μ_n into (6.9), the resistance is obtained as function of V_{BD}. This holds directly for the case of NPT. As discussed above, a slight reduction of R_Ω is possible by PT-layout. To obtain the minimal resistance attainable the NPT-value is to be multiplied by a factor which is nearly independent of the blocking voltage, see Sect. 6.4. Calculations like those for Fig. 6.4 using $n = 8.03$ yield that this factor is $0.868 \pm 1\%$ for the whole voltage range above 500 V. Hence multiplying (6.10) with this factor and inserting the mentioned n-value one obtains for the minimal resistance of SiC-devices:

$$R_{\Omega,\min} = 0.87 \cdot \frac{2^{2.43} \cdot \left(\frac{B}{9.03}\right)^{0.43} \cdot V_{BD}^{2.43}}{\mu_n \cdot \varepsilon \cdot A} \tag{6.16}$$

The minimal resistance $R_{\Omega,min}$ obtained in this way is shown in Fig. 6.8 as function of V_{BD} (using the relative dielectric constant $\varepsilon_r = 9.66$). For comparison also the equal dependency for silicon is depicted. The curves are approximately straight lines on this double-logarithmic scale. The resistance for SiC is at 1200 V by a factor 740 lower than for Si, the blocking voltage for a given R_Ω is well a decade higher. The allowed field has been assumed in the calculations to be equal to E_c. The curves represent hence lowermost limits for R_Ω which are referred to as the silicon carbide respectively silicon 'unipolar limits'. As shown by the experimental points depicted in the figure, practical SiC-devices do not reach this limit. The resistance of the purely unipolar Schottky diode taken from [Pet01] lies an order above the theoretical limit.

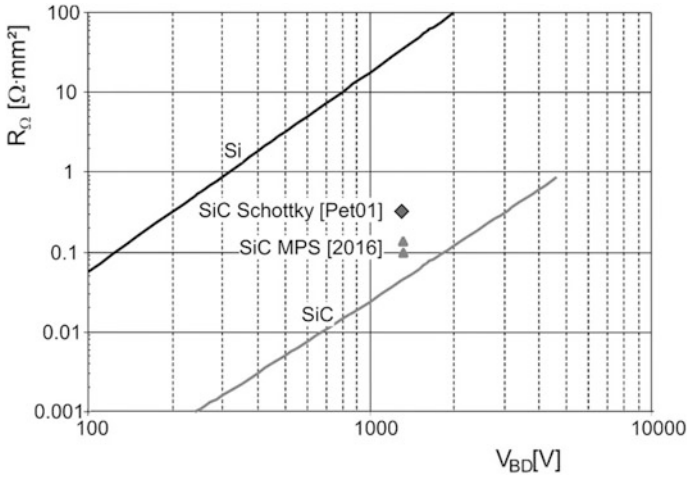


Fig. 6.8 Unipolar resistance $R_{\Omega,\ min}$ for Si and SiC as function of the breakdown voltage. Measurements from [Pet01] and actual MPS diodes from 2016 production are indicated for comparison

The graphs for the "unipolar limit" vary in the literature by some small amount. Possibly some authors use E_c as constant and do not consider its doping dependency. The field shape and doping are taken into account here by using the respectively approach for the ionization rates for Si and for SiC.

Figure 6.9, taken from [Pet01], shows the IV characteristics of a SiC Schottky diode rated for 1200 V with 10 mm^2 active area. The characteristics for 25 and 125 °C are shown. The forward characteristic has a resistive behavior, the resistance extracted from the characteristic at 25 °C is depicted in Fig. 6.8 as [Pet01]. The threshold voltage at 25 °C is in agreement with Fig. 6.6. With increasing temperature, the on-resistance increases in accordance with the decrease of the electron mobility. In reverse direction the current shows a significant increase already appreciably below the breakdown voltage, i.e. the characteristics are soft, and the soft behavior increases with temperature. The softness of the characteristic limits the usable blocking voltage which decreases with increasing temperature. In theory the breakdown voltage should increase with the temperature due to the decreasing avalanche coefficients and increasing critical field strength. The softness is probably mainly caused by lattice defects and/or surface defects. An additional cause is the image force introduced by Eqs. (6.5) and (6.6). Because of the high electric field this results in a significant reduction of the Schottky barrier in SiC-devices. For a field of 1×10^6 V/cm Eq. (6.6) yields $\Delta\Phi = 0.122$ V, which according to (6.5) results in an enhancement of the saturation current at 300 K by the factor 112, at 400 K by a factor 34.4.

The charge necessary for a given on-conductance is extracted (to a large part) at commutation into the reverse direction and appears as reverse current integral. The resistance and the charge $Q = q \cdot N_D \cdot A \cdot w_B$ are connected by

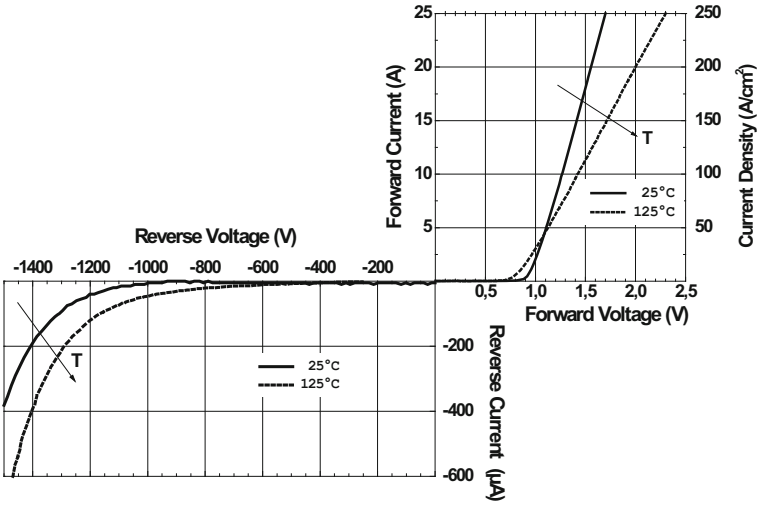


Fig. 6.9 IV-characteristic of a 1200-V SiC Schottky diode with 10 mm^2 active area at 25 °C and at 125 °C. Figure taken from [Pet01]

$$G = \frac{1}{R_\Omega} = \frac{q\mu_n N_D A}{w_B} = \frac{\mu_n Q}{w_B^2} \tag{6.16}$$

The inverse relationship between the on-resistance and the stored charge is similar to pin-diodes. In contrast to the latter the charge of a Schottky-diode does not increase with temperature, which is advantageous with respect to dynamical properties, but disadvantageous with respect to the forward voltage. For SiC devices with their small base width the charge is very small for a given R_Ω. Nevertheless it can be advisable in dimensioning of SiC Schottky-diodes to make the resistance R_Ω not *unnecessarily* small and the charge Q accordingly high. We will come back to the dynamical behavior of SiC-Schottky diodes in Sect. 6.5.3.

6.5.2 Merged Pin Schottky (MPS) Diodes

To reduce the influence the image force and improve the surge current capability, merged pin-Schottky structures (MPS-diodes) have been introduced [Bjo06, Rup06, Sin00]. Most Schottky diodes already contain a p-zone at the edge of the active area for field termination as shown in Fig. 6.10a. In Fig. 6.10b additional high p-doped areas are implemented, which act as p-emitters and inject carriers if the forward voltage drop is above the junction voltage of the SiC pin-diode. This is the case in the range of 2.8 V, as discussed above. An additional solution is the implementation of p-layers in analogy with silicon merged pin-Schottky (MPS) diodes as shown in Fig. 5.28. Such a structure is shown in Fig. 6.10c. These p-layers are separated by a relatively small distance of n^--material with the aim to completely shield the Schottky contact from the high electric field in reverse bias.

The surge-current behavior of these structures was investigated in [Hei08c]. The I–V characteristic of diodes with the three structures in Fig. 6.10 is shown in Fig. 6.11. The structure with p^+-regions (b) shows the lowest forward voltage drop for high current. The effect of the pin-diode is clearly noticeable for currents above

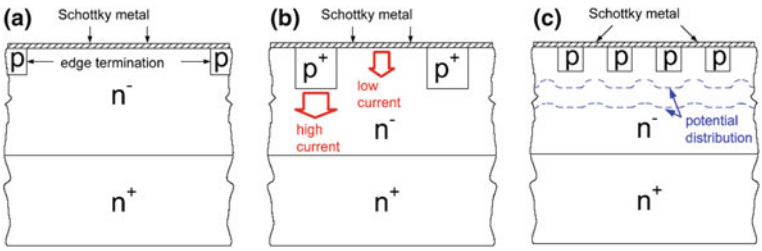


Fig. 6.10 Structure of **a** conventional SiC Schottky-diode, **b** MPS diode structure with p^+-regions to improve the surge-current capability [Bjo06] and **c** MPS diode structure with p-regions to shield the Schottky junction from high electric fields [Sin00]. Figure similar to [Hei08b]

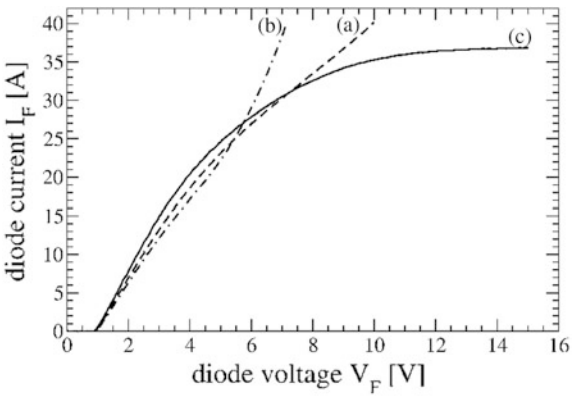


Fig. 6.11 Forward IV-characteristic for SiC diodes for the three structures shown in Fig. 6.10. Specified voltage 600 V, rated current 4 A for all three diodes. Figure similar to [Hei08b]

20 A, when the p-layers inject current. This structure shows a high surge-current capability. A similar effect, although not as strong, can also be seen for structure (a), where the pin-contribution caused by the edge region injects carriers. Structure (c) shows no pin-behavior at all. Although it contains p-layers, they do not inject carriers and it remains unipolar. Moreover, the I–V characteristic deviates from the ohmic behavior at currents above 20 A. At high forward voltage, a significant electric field arises across the middle layer, which has a thickness of only some µm. The increased field will then cause the electron mobility to decrease and will result in an increased forward voltage drop.

A further significant contribution is the temperature increase during measurement. Therefore, the surge-current capability is limited for this structure. However, a structure as displayed in Fig. 6.10c is appropriate to shield the Schottky contact from high electric fields. The highest electric field occurs at the edges of the p-zones. The surface between them is released. Therefore, this structure avoids the reverse bias leakage current of the Schottky junction.

Figure 6.12 shows a top view of the arrangement of p-regions and Schottky zones of a recent MPS-diode [Rup12, Rup14]. The Schottky regions are formed as hexagons, with a pin-region in the middle, with some larger-area pin-regions and with a pin-region of the largest size at the edge of the active area. The total Schottky area is about 50%. The structure combines the advantages of Fig. 6.11b, c.

SiC MPS diodes for 1200 V demonstrate a kind of triggering of the bipolar part at surge-current conditions. Figure 6.13 shows a measurement and electro-thermal simulation of a 1200 V SiC MPS diode with 10A rated current. The current density at rated current is 340 A/cm^2. The simulation simplifies a 10 µm wide 2-dimensional structure with 50% Schottky-area and 10 µm extension since simulation of the complex 3D-structure in Fig. 6.12 is due to extensive computing time, which is not possible, yet.

In the measurement, close to the marked point I, the p-regions start to inject and a branch with negative differential resistance starts to form. In the simulation with a simplified 2D-stucture, this is a pronounced triggering point. The device strongly

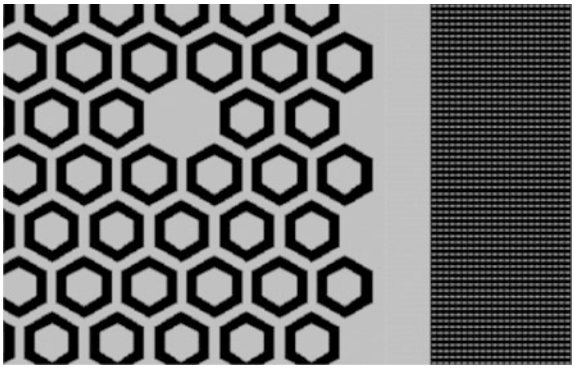


Fig. 6.12 Top view of the structure of an MPS-diode. Black: Schottky regions. Grey: pin-regions. Dotted area on the right side: Junction termination. Figure from R. Rupp, Infineon

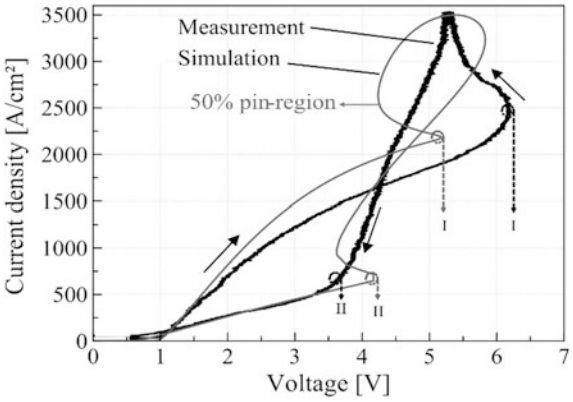


Fig. 6.13 Surge-current pulse, half-sinus 10 ms applied to a 1200 V SiC MPS diode with 10 A rated current, T = 25 °C, measurement and simulation. © 2016 IEEE. Reprinted, with permission, from [Pal16]

heats up and reaches up to 800 K. At high temperature the mobilities strongly decrease, and the descending branch follows another line. In simulation the mobility model with the temperature dependent parameters from [Scr94] is used. The extinction point at II is pronounced in simulation, in the measurement it is blurred due to the more complex structure with different sizes of p-layers.

The behavior of MPS diodes strongly depends on the device geometry, e.g. the size of the pin-area relative to the Schottky area [Fic14, Pal16]. The triggering of the minority carrier injection can be approximated if it is considered that in the Schottky-mode the electron current must flow laterally below the p-layer, see Fig. 6.14.

The voltage drop of lateral current flow across the length L can be calculated according to [Ogu04] as

$$V_p = \int_0^L R \cdot j_p \cdot y \cdot dy = \frac{1}{2} \cdot R \cdot j_p \cdot L^2 \tag{6.17}$$

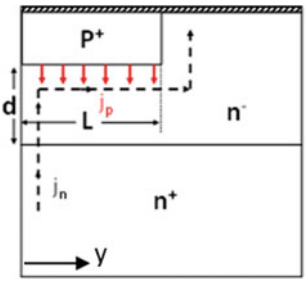


Fig. 6.14 Simplified drawing for the triggering state in an MPS diode. © 2016 IEEE. Reprinted, with permission, from [Pal16]

Here R is the sheet resistivity in $\Omega/\square$ and L is the lateral length of the pin-region. R can be approximated by ρ/d for a not too thin epi-layer thickness d, $d > ½\ L$. If it is assumed that the p-layers inject as V_p is equal to the difference in built-in voltage of the pn-junction and threshold of the Schottky-junction $V_{bi} - V_S$, this results in [Pal16]

$$j_p = \frac{2 \cdot (V_{bi} - V_S)}{R \cdot L^2} \tag{6.18}$$

Since $\rho = 1/(q \cdot \mu_n \cdot N_D)$ contains the bulk mobility μ_n in SiC which decreases strongly with temperature, this explains the triggering at lower current for the high temperature branch in Fig. 6.13. With these simplifications one can approximate the triggering state with the size of the pin-area in an MPS diode. For the structure in Fig. 6.12, it was found that p-areas start to contribute subsequently to the current conduction: first the pin-region at the edge termination triggers, followed by the additionally large p^+ cells, as last the small p^+ cells contribute [Rup12].

Further considerations on the triggering of the p^+-injection are reported in [Hua16], an extended model is given in [Niw17]. Work on high-voltage MPS diodes is presented in [Niw17]. The devices are fabricated with an epitaxial p^+-anode layer, and intensively investigated is the "snapback" phenomenon which is visible in the measurement in Fig. 6.13 at 6.2 V. In [Niw17] p-layers with wide L (see Fig. 6.13) of 50–150 µm are used, and for L, in case it is large enough, the transition from the Schottky-region the pin-operation becomes smooth. As design guideline of snapback-free hybrid operating MPS diodes, $L/d > 1.5$ is given. MPS diodes with breakdown voltages up to 11.3 kV are shown.

SiC MPS diodes have shown a high capability to withstand stress in avalanche, as shown for 600 V in [Rup06] and 1200 V in [Rup14]. The key enabler is the intentional field crowding at the bottom of the implanted p-layers in the cell structure, and then avalanche breakdown occurs at the edges of the p-layers (see Fig. 6.10c). Additionally, it is necessary to balance the breakdown in the junction termination with the breakdown in the cell field and to design the cell field in a way to have a lower breakdown voltage than the junction termination [Dra15].

For MPS diodes with such design, an increase of the breakdown voltage with temperature is found. A linear increase of V_{BD} with 0.35 V/K is measured, see Fig. 6.15. This differs from the Schottky-diode in Fig. 6.9. Obviously, the effect of

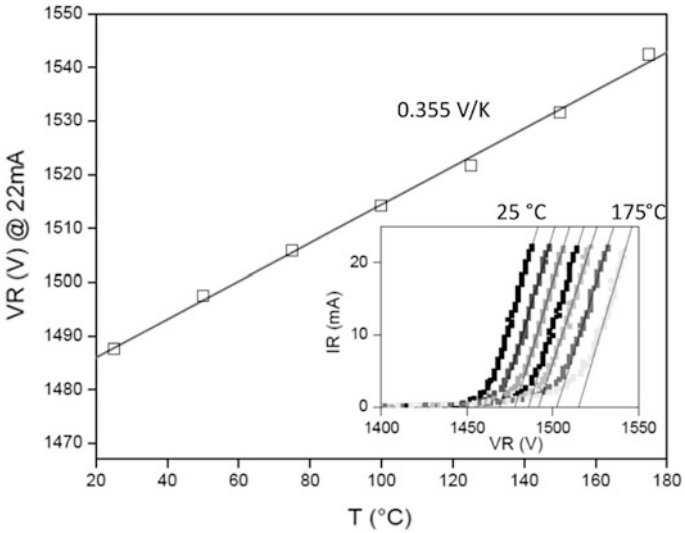


Fig. 6.15 Temperature dependency of the breakdown voltage in a 1200 V MPS diode. Figure from R. Rupp, Infineon

increased leakage current by the image force is reduced if the Schottky junction is released from high electric field. This can be done by a narrow distance of the p-layers in MPS diodes.

6.5.3 *Switching Behavior and Ruggedness of SiC Schottky and MPS Diodes*

SiC-Schottky-diodes are in competition with fast silicon pin-diodes, they can gain a market by a better switching behavior. A great advantage of SiC-diodes is their much smaller charge necessary for a given on-state conductance G, since this charge appears during commutation as reverse current integral. In a chopper circuit the reverse current pulse of the diode during turn-on of the switching element adds to the forward current in the latter, hence the lower charge leads to reduced turn-on losses in the switch.

For the NPT-case (triangular field) the ratio between the charge and conductance is

$$\frac{Q}{G} = QR_{\Omega} = \frac{qN_D A w_B}{q\mu_n N_D A / w_B} = \frac{w_B^2}{\mu_n} = \frac{4V_{BD}^2}{\mu_n \mathsf{E}_c^2} \tag{6.19}$$

The figure-of-merit of a semiconductor for a high on-state conductance per stored charge is hence $\mu_n E_c^2$ [Sco91], a figure which includes static *and* dynamical properties. In a form considering also the mobility of holes in pin-diodes (μ_n replaced by $\mu_n + \mu_p$) it can be used for an approximate comparison of SiC Schottky diodes with Si pin-diodes. Assuming a factor 10 in the critical field and a factor 1/2 from mobilities, the product $Q{\cdot}R_\Omega$ of a 4H-SiC Schottky-diode is obtained to be a factor of 50 smaller than in a Si pin-diode. Hence for equal resistance of the base region the *charge* in the SiC-diode is by the same factor smaller than the stored charge in the Si pin-diode. Whereas the product $Q{\cdot}R_\Omega$ for a given material and breakdown voltage has a fixed value, the factors Q and R_Ω can be changed (in an inverse manner) via the active area A of the device. A lower limit to the area is set up by the losses to be conducted away.

A disadvantage of power Schottky diodes are oscillations of reverse voltage and current produced after commutation by the capacitance of the Schottky junction C_j and the stray inductance L_p, as shown in [Sco91]. If only Si is replaced by SiC in high-current applications, huge problems with oscillations may occur [Win12, Lut14]. The capacitance of the SiC-device is given by

$$C_j = \sqrt{\frac{q \cdot \varepsilon \cdot N_D}{2 \cdot (V_j + V_r)}} \cdot A \tag{6.20}$$

and has its maximum C_j (0 V) when the applied voltage V_r is zero. Due to the two decades higher doping N_D it is one decade higher than in the case of a Si-diode with the same active area. A tail current which occurs in Si bipolar devices and which damps oscillations is missing with SiC. The oscillations can be damped making the oscillation period $\tau_{osc} \sim \sqrt{L_p}$ not larger than the turn-on time of the switching element in the chopper circuit, which time should be small also, however. This is a challenge for a new packaging technology which requires much lower parasitic inductance. There is much effort invested today in the development of packages with very low inductance. In a package rated for 400 A with L_p of only 1.4 nH, the oscillations are strongly reduced [Bec16]. Special care has to be taken in case of paralleling, since non-symmetric inductivities in front of parallel connected chips will lead to bad dynamic current sharing and internal LC-oscillations.

MPS diodes show a high dynamic ruggedness at turn-off under overload conditions. Measurements of a 5 A MPS diode in a double-pulse circuit according to Fig, 5.19 executed with an IGBT rated to 90 A and 1200 V at R_G = 20 Ω are displayed in Fig. 6.16 [Fic15]. Up to a current of 25 A the diode shows the typical fast-switching characteristics of a Schottky diode with no bipolar charge, and the occurring reverse recovery charge is only caused by the space charge capacity and independent from the forward current. At 50 A, however, the tendency to a bipolar reverse-recovery behavior can be presumed. At this forward current, the first p-doped regions are already active and the diode is slightly bipolar. When imposing a forward current of 75 A before the turn-off, the diode becomes bipolar. This is reflected in the switching behavior, which is similar to the behavior of a pin-diode.

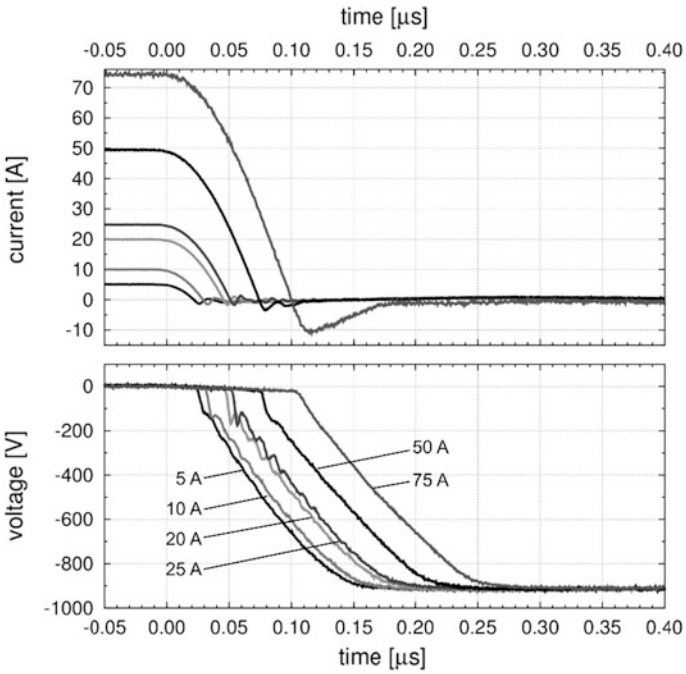


Fig. 6.16 Turn-off behavior of an MPS diode with rated current 5 A, $T = 25$ °C. The forward current varied between one to fifteen times the rated current. © 2015 Elsevier. Reprinted from [Fic15] with permission from Elsevier

The diodes were able to withstand turn-off at high *di*/*dt* and *dv*/*dt* executed by setting $R_G = 1\ \Omega$ at more than fifteen times the rated current. Above approximately eight to ten times the rated current, their switching behavior equals that of a pin-diode.

SiC MPS diodes have become mature devices. They are used in the 600 V range in combination with Silicon MOSFETs for high-frequency applications in Power Factor Correction (PFC) circuits and in DC-DC converters. In application as freewheeling diodes for IGBTs, SiC MPS diodes allow to reduce turn-on losses of the IGBT due to the very low reverse recovery charge, see Sect. 5.7.2.

SiC Schottky diodes are preferred for several applications that involve high switching frequencies. SiC devices are still young and, when compare them to Si devices, one must remember that the silicon technology has been improved for several decades to reach the device performance achieved today. For silicon carbide there still is a high potential for optimization of crystal quality as well as for processing technology and device design. It must be considered, however, that SiC is a challenge for the packaging technology because of the higher stiffness compared to Si, respectively the higher Young's modulus of SiC compared to Si.

Therefore, replacements of Si by SiC must consider reliability, especially power cycling capability (see Chap. 12 on reliability).

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